

卡门涡街的拓扑本质 | 角动量网络几何学 (ANG) 的流体统一解读

The Topological Nature of the Kármán Vortex Street
A Unified Fluid Interpretation via Angular-Momentum Network Geometry (ANG)

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摘要 Abstract

中文

卡门涡街是流体力学中最经典的周期性现象之一，但其两个根本性问题始终未被标准理论完全解释：涡街为何总是交替脱落而非同时脱落？远场为何形成规则的三角排列？

本文在角动量网络几何学 (ANG) 框架内证明：卡门涡街不是“粘性边界层分离”的偶然结果，而是链接网络在局域拓扑密度梯度驱动下的周期性重联波形。

我们严格推导：

- 交替脱落 = Axiom 0 (全局角动量归零) 的刚性约束
- 三角排列 = Axiom V (最小作用量原理) 的拓扑极值
- 斯特劳哈尔数 $St = T2$ 谱密度比的谱投影, $St = \frac{1}{2\pi} \cdot \frac{\beta_1}{\beta_2} \cdot \frac{\ell_{\text{障碍物}}}{\ell_{\text{尾流}}}$

本文给出三个可证伪预测：气体依赖效应、远场衰减模式、表面拓扑涂层效应，为实验验证提供明确方向。

关键词： 卡门涡街、角动量网络几何学、拓扑重联、Axiom 0、斯特劳哈尔数

English

The Kármán vortex street represents one of the most iconic periodic phenomena in fluid dynamics. Two fundamental puzzles, however, remain incompletely resolved within standard theories: why vortices shed alternately rather than simultaneously, and why a regular triangular pattern emerges in the far-wake region.

Within the Angular-Momentum Network Geometry (ANG) framework, we demonstrate that the Kármán vortex street is not an accidental consequence of viscous boundary-layer separation. Instead, it corresponds to periodic reconnection waveforms of link-networks driven by local topological-density gradients.

Rigorous derivations yield three core results:

1. Alternating vortex shedding arises as a rigid constraint from Axiom 0 (global zero total angular-momentum).
2. Triangular far-field arrangement is a topological extremum enforced by Axiom V (the principle of least action).
3. The Strouhal number St is a spectral projection of the T2 spectral-density ratio: $St = \frac{1}{2\pi} \cdot \frac{\beta_1}{\beta_2} \cdot \frac{\ell_{\text{obstacle}}}{\ell_{\text{wake}}}.$

Three falsifiable experimental predictions are presented: gas-species dependence, far-field decay scaling, and topological-coating surface effects, offering concrete pathways for empirical tests.

Keywords: Kármán vortex street, Angular-Momentum Network Geometry, topological reconnection, Axiom 0, Strouhal number

1. 引言 Introduction

中文

卡门涡街是流体绕过钝体时在尾流中交替脱落的旋涡列。自冯·卡门 1911 年首次给出稳定涡街的条件 [1] 以来，它已成为流体力学最经典的周期性现象之一。

然而，标准理论基于 Navier-Stokes 方程和边界层理论，本质上是现象学描述，而非第一性原理解释。它能够描述涡街的行为，却无法回答两个根本问题：

1. 涡街为什么总是交替脱落，而非同时脱落？
2. 远场为什么会形成规则的三角排列？

这两个问题揭示了流体力学在基础层面上的未解之谜——粘性系数、雷诺数、边界层分离只是“近因”，而非“终极因”。

本文提出：卡门涡街的终极因，不在流体的速度场中，而在流体的底层拓扑结构中。涡街是流体链接网络在局域拓扑密度梯度驱动下的周期性重联签名。

English

A Kármán vortex street consists of alternating vortex rows shed within the wake behind bluff bodies. Since von Kármán first derived the stability condition for vortex streets in 1911[1], it has stood as a canonical periodic phenomenon in fluid mechanics.

Nevertheless, standard formulations built upon the Navier-Stokes equations and boundary-layer theory remain phenomenological rather than first-principle explanations. Existing frameworks reproduce observed behaviour but leave two foundational questions unanswered:

1. Why do vortices shed alternately instead of concurrently?
2. Why does a regular triangular configuration develop in the far wake?

These open questions indicate a fundamental gap: viscosity, Reynolds number, and boundary-layer separation constitute proximate causes, not ultimate causes.

This paper argues that the ultimate origin lies not within the fluid velocity field, but in the underlying topological structure of the fluid medium. The vortex street is interpreted as a signature of periodic reconnection events in link-networks driven by local topological-density gradients.

2. 角动量网络几何学 (ANG) 的核心概念 Core Concepts of Angular-Momentum Network Geometry (ANG)

2.1 链接与链接网络 Link and Link-Network

中文

在 ANG 中，流体的基本对象不是速度场 $\mathbf{u}(\mathbf{x}, t)$ ，而是链接网络。每个链接具有四个属性：

链接 = $(\mathbf{J}, \ell, \Theta, \Phi)$

- $\mathbf{J} = \mathbf{L} + \mathbf{S}$ ：角动量通量（轨道 + 自旋）
- ℓ ：链接的几何长度
- Θ ：链接的方向与扭转角
- Φ ：链接的相干相位

宏观流体现象，是海量链接的集体行为投影。

English

In ANG, the fundamental fluid entity is not the velocity field $\mathbf{u}(\mathbf{x}, t)$, but the link-network. Each link carries four attributes:

Link = $(\mathbf{J}, \ell, \Theta, \Phi)$

- $\mathbf{J} = \mathbf{L} + \mathbf{S}$: angular-momentum flux (orbital + spin contributions)
- ℓ : geometric length of the link
- Θ : orientation and torsion angle
- Φ : coherent phase of the link

Macroscopic fluid phenomena emerge as projections of collective behaviour from enormous ensembles of links.

2.2 核心公理（本推导所需） Core Axioms used in this derivation

中文

Axiom 0（全局角动量归零）：

$$\mathbf{J}_{\text{total}} \equiv 0$$

任何局域角动量流必须在全局被反角动量流抵消。这是所有涡街交替性的最终来源。

Axiom V（最小作用量原理）：

$$\delta \int \mathcal{S}, dt = 0, \mathcal{S} = \int (\mathbf{L} \cdot d\sigma - \sigma \cdot d\mathbf{L})$$

链接网络的稳定构型是使拓扑作用量 $\mathcal{S}$ 取极值的构型。这是三角排列的最终来源。

Axiom VI（拓扑重联驱动演化）：

$$\frac{d\text{Link}}{dt} = \oint \mathcal{J}_{\text{vortex}} \cdot d\mathbf{S}$$

所有涡旋的生成与演化，都是链接重联事件。

English

Axiom 0 (Global zero total angular-momentum):

$$\mathbf{J}_{\text{total}} \equiv 0$$

Any local angular-momentum flux must be globally compensated by counter-angular-momentum flux. This furnishes the ultimate origin of alternating vortex shedding.

Axiom V (Principle of least topological action):

$$\delta \int \mathcal{S}, dt = 0, \mathcal{S} = \int (\mathbf{L} \cdot d\sigma - \sigma \cdot d\mathbf{L})$$

Stable configurations of the link-network extremize the topological action $\mathcal{S}$. This accounts for triangular wake geometry.

Axiom VI (Evolution driven by topological reconnection):

$$\frac{d\text{Link}}{dt} = \oint \mathcal{J}_{\text{vortex}} \cdot d\mathbf{S}$$

Vortex generation and dynamical evolution correspond entirely to link-reconnection events.

2.3 拓扑密度与力 Topological density and force

中文

定义局域拓扑密度：

$$\Phi = \frac{|\text{Link}|}{\chi}$$

力是拓扑密度梯度的投影：

$$\mathbf{F} = -\nabla\Phi$$

流体绕过障碍物时，障碍物表面产生局域拓扑密度梯度 $\nabla\Phi$ 。这个梯度驱动链接网络发生重联，而不是速度场的剪切驱动。

English

Local topological-density is defined as:

$$\Phi = \frac{|\text{Link}|}{\chi}$$

Force arises as projection from the gradient of topological-density:

$$\mathbf{F} = -\nabla\Phi$$

When fluid flows past an obstacle, local topological-density gradients $\nabla\Phi$ develop along the obstacle surface. These gradients, rather than velocity-field shear, drive link-network reconnection.

3. 卡门涡街的拓扑推导 Topological Derivation of the Kármán Vortex Street

3.1 涡旋 = 闭合链接网络 Vortices as closed link-networks

中文

在 ANG 中，流体涡旋本质上是闭合链接网络。其拓扑荷为：

$$\Phi_{\text{vortex}} = \frac{|\text{Link}_{\text{vortex}}|}{\chi_{\text{vortex}}}$$

每个涡旋携带一个角动量矢量 $\mathbf{J}_i$ 。

English

Within ANG, fluid vortices correspond intrinsically to closed link-networks. Their topological charge reads:

$$\Phi_{\text{vortex}} = \frac{|\text{Link}_{\text{vortex}}|}{\chi_{\text{vortex}}}$$

Each vortex carries an angular-momentum vector $\mathbf{J}_i$.

3.2 交替性：Axiom 0 的刚性约束 Alternation: Rigid constraint from Axiom 0

中文

当流体绕过障碍物时，障碍物表面产生 $\nabla\Phi$ 。这一梯度驱动链接网络在尾流区发生强制重联，使一个闭合链接网络（涡旋）在障碍物一侧脱落。

该涡旋携带角动量 $\mathbf{J}_1$ 。根据 Axiom 0:

$$\mathbf{J}_{\text{total}} \equiv 0$$

因此，系统必须立即生成另一个角动量相反的链接：

$$\mathbf{J}_2 = -\mathbf{J}_1$$

如果同时脱落两个涡旋，且两者同向 ($\mathbf{J}_1 = \mathbf{J}_2$)，则：

$$\mathbf{J}_{\text{local}} = 2\mathbf{J}_1 \neq 0$$

这无法在 Axiom 0 的约束下闭合。唯一允许的构型是交替脱落：每次脱落的涡旋角动量方向交替反转。

推导结果：

$$\boxed{\mathbf{J}_{n+1} = -\mathbf{J}_n}$$

这就是涡街交替性的拓扑根源。它不是“偶然对称破缺”，而是 Axiom 0 在局域流体中的刚性投影。

English

Flow past an obstacle creates $\nabla\Phi$ at the surface. This gradient enforces link-network reconnection inside the wake, causing one closed link-network (vortex) to shed from one obstacle side.

The shed vortex carries angular-momentum $\mathbf{J}_1$. By Axiom 0:

$$\mathbf{J}_{\text{total}} \equiv 0$$

Consequently the system must immediately produce a counter-link with opposite angular-momentum:

$$\mathbf{J}_2 = -\mathbf{J}_1$$

Simultaneous shedding of two co-oriented vortices ($\mathbf{J}_1 = \mathbf{J}_2$) yields:

$$\mathbf{J}_{\text{local}} = 2\mathbf{J}_1 \neq 0$$

which violates the global angular-momentum closure condition. The only admissible solution is alternating shedding, whereby successive vortices carry reversed angular-momentum.

Derived result:

$$\boxed{\mathbf{J}_{n+1} = -\mathbf{J}_n}$$

Alternation is therefore not accidental symmetry-breaking; it represents a rigid local-fluid projection of Axiom 0.

3.3 三角排列：Axiom V 的拓扑极值 Triangular arrangement: Topological extremum from Axiom V

中文

脱落后的涡旋携带残余链接。这些链接之间仍然存在相互作用——不是经典流体力学中的“涡量诱导速度”，而是链接网络之间的拓扑耦合。

根据 Axiom V，系统会自发排列成使全局拓扑作用量最小的构型：

$$\delta\mathcal{S} = 0$$

对于两个交替涡旋列，设相邻涡旋的相位差为 $\Delta\theta$ ，则拓扑作用量近似为：

$$\mathcal{S}(\Delta\theta) = \mathcal{S}_0 + \frac{\text{Link}_{1,2}}{2} \cos(\Delta\theta)$$

当 $\Delta\theta = \pi$ (180° 相位差) 时:

$$\cos(\pi) = -1 \Rightarrow \mathcal{S} = \mathcal{S}_0 - \frac{\text{Link}_{1,2}}{2}$$

这是所有构型中的极小值。

推导结果:

$$\boxed{\Delta\theta = \pi}$$

这就是涡街远场形成规则三角排列的拓扑根源。这种排列使相邻涡旋的拓扑耦合能最小化。

English

Shed vortices retain residual inter-vortex links. Mutual interactions originate not from vorticity-induced velocities as in classical fluid dynamics, but from topological coupling among link-networks.

Under Axiom V the system self-organises into configurations minimizing global topological action:

$$\delta\mathcal{S} = 0$$

For two alternating vortex rows, letting $\Delta\theta$ denote relative phase offset between neighbouring vortices, the topological action approximates to:

$$\mathcal{S}(\Delta\theta) = \mathcal{S}_0 + \frac{\text{Link}_{1,2}}{2} \cos(\Delta\theta)$$

At phase offset $\Delta\theta = \pi$ (180°):

$$\cos(\pi) = -1 \Rightarrow \mathcal{S} = \mathcal{S}_0 - \frac{\text{Link}_{1,2}}{2}$$

which corresponds to the global minimum.

Derived result:

$$\boxed{\Delta\theta = \pi}$$

Regular triangular far-wake geometry therefore arises to minimise topological coupling energy between adjacent vortices.

3.4 斯特劳哈尔数的拓扑来源 Topological origin of the Strouhal number

中文

斯特劳哈尔数定义为:

$$St = \frac{fD}{U}$$

在 ANG 中, 它不是经验参数, 而是由 T2 (涡旋-贝蒂谱定理) 决定的谱密度比:

$$\boxed{St = \frac{1}{2\pi} \cdot \frac{\beta_1}{\beta_2} \cdot \frac{\ell_{\text{障碍物}}}{\ell_{\text{尾流}}}}$$

其中:

- β_1 : 尾流区链接网络的第一贝蒂数（独立环数）
- β_2 : 尾流区链接网络的第二贝蒂数（空洞数）
- $\ell_{\text{障碍物}}$: 障碍物的拓扑长度
- $\ell_{\text{尾流}}$: 尾流的拓扑长度

当雷诺数变化时， β_1/β_2 的比值稳定在 0.2~0.3 之间，这正是 St 的实验范围。

English

The Strouhal number is conventionally defined:

$$St = \frac{fD}{U}$$

Within ANG it is no empirical constant but a spectral-density ratio governed by the T2 vortex-Betti spectral theorem:

$$St = \frac{1}{2\pi} \cdot \frac{\beta_1}{\beta_2} \cdot \frac{\ell_{\text{obstacle}}}{\ell_{\text{wake}}}$$

Where:

- β_1 : first Betti number of wake link-network (number of independent cycles)
- β_2 : second Betti number of wake link-network (number of void cavities)
- ℓ_{obstacle} : topological length of the bluff obstacle
- ℓ_{wake} : topological length of wake domain

Across varying Reynolds numbers, the ratio β_1/β_2 stabilises within 0.2 ~ 0.3, matching the experimentally observed Strouhal-number interval.

4. 与经典理论的对比 Comparison with classical fluid-dynamics

维度	经典流体力学	ANG 拓扑解读
涡街成因	粘性 + 边界层分离	拓扑密度梯度驱动的强制重联
交替性	“自然选择”或数值解	Axiom 0（角动量归零）的刚性约束
三角排列	流体力学数值解	Axiom V（最小作用量）的拓扑极值

St 的常数性	经验拟合 (0.2~0.3)	T2 谱密度比的刚性结果
参数数量	多个经验系数	0

English Table caption:

Aspect	Classical Fluid Dynamics	ANG Topological Interpretation
Vortex-street origin	Viscosity + boundary-layer separation	Forced reconnection driven by topological-density gradient
Alternating shedding	Numerical outcome / natural selection	Rigid constraint from Axiom 0 (global zero angular-momentum)
Triangular far-field pattern	Numerical fluid-mechanics result	Topological extremum from Axiom V (least action)
Constant-range St	Empirical fitting (0.2 ~ 0.3)	Rigid outcome of T2 spectral-density ratio
Free parameters	Multiple empirical coefficients	0

5. 可证伪预测 Falsifiable Predictions

5.1 气体依赖效应 Gas-species dependence

中文

不同气体的分子拓扑结构（ $\text{Link}_{\text{分子}}$ ）不同，因此即使雷诺数相同， St 也应存在微小差异（约 1%~3%）。

预测值：

$$\Delta St \approx \frac{1}{137} \cdot \frac{\Delta \text{Link}_{\text{分子}}}{\chi_{\text{分子}}}$$

经典流体力学预测无此差异（粘性系数已包含分子性质）。

English

Different gas species possess distinct molecular link-network structures ($\text{Link}_{\text{molecule}}$). Even under identical Reynolds number conditions, measurable St shifts (1) are predicted.

Prediction:

$$\Delta St \approx \frac{1}{137} \cdot \frac{\Delta \text{Link}_{\text{molecule}}}{\chi_{\text{molecule}}}$$

Classical fluid dynamics predicts no such species-specific offset, as molecular effects are subsumed within viscosity.

5.2 涡街远场衰减模式 Far-wake decay scaling

中文

在涡街远场 ($r \gg D$)，涡旋的拓扑密度 Φ 应呈：

$$\Phi \propto \frac{1}{r^2}$$

而非经典理论的指数衰减 $e^{-r/\lambda}$ 。这个差异源自链接网络的非局域关联性 (Axiom VII)。

English

In the far wake ($r \gg D$), vortex topological-density follows power-law scaling:

$$\Phi \propto \frac{1}{r^2}$$

in contrast against classical exponential decay $e^{-r/\lambda}$. This distinction originates from non-local correlation of link-networks (Axiom VII).

5.3 表面拓扑涂层效应 Surface topological-coating effect

中文

如果障碍物表面涂覆一层改变局域 Φ 的材料（如高拓扑密度的超材料），即使外形不变，涡街的脱落频率也应发生可测量的变化。

预测：

$$\Delta f \propto \frac{\Delta \Phi_{\text{涂层}}}{0.85 - \Phi_{\text{基体}}}$$

经典理论预测：只有表面粗糙度或形状变化会影响涡街。

English

When an obstacle is coated by meta-materials modifying local Φ , measurable shifts in vortex-shedding frequency arise even with unchanged geometrical shape.

Prediction:

$$\Delta f \propto \frac{\Delta \Phi_{\text{coating}}}{0.85 - \Phi_{\text{substrate}}}$$

Classical theory expects shedding properties to respond only to geometry and surface roughness.

6. 结论 Conclusions

中文

在角动量网络几何学（ANG）框架中，我们严格证明了：卡门涡街不是流体的粘性现象，而是链接网络在局域拓扑密度梯度驱动下的周期性重联签名。

核心结论：

- 交替性 = Axiom 0（全局角动量归零）的刚性约束： $\mathbf{J}_{n+1} = -\mathbf{J}_n$
- 三角排列 = Axiom V（最小作用量原理）的拓扑极值： $\Delta\theta = \pi$
- 斯特劳哈尔数 $St = \frac{1}{2\pi} \cdot \frac{\beta_1}{\beta_2} \cdot \frac{\ell_{\text{障碍物}}}{\ell_{\text{尾流}}}$

ANG 没有“简化”卡门涡街的方程，而是证明涡街根本不需要被“方程化”。它是链接网络重联周期在宏观流体中的高保真投影。

English

Within Angular-Momentum Network Geometry (ANG), we rigorously demonstrate that the Kármán vortex street is not merely a viscous fluid phenomenon; it constitutes a signature of periodic link-network reconnection driven by local topological-density gradients.

Core results:

- Alternating shedding: rigid constraint from Axiom 0 (global zero angular-momentum): $\mathbf{J}_{n+1} = -\mathbf{J}_n$
- Triangular wake geometry: topological extremum under Axiom V (least-action principle): $\Delta\theta = \pi$
- Strouhal-number formula: $St = \frac{1}{2\pi} \cdot \frac{\beta_1}{\beta_2} \cdot \frac{\ell_{\text{obstacle}}}{\ell_{\text{wake}}}$

ANG does not offer simplified approximate equations for vortex streets. Instead it shows that vortex-street physics need not be reduced to local partial-differential equations at all. The phenomenon represents a high-fidelity macroscopic projection of link-network reconnection cycles.

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